Problem 05: Karachi Food Festival

Time limit: 2 seconds

The annual Karachi Food Festival has ***N*** unique food stalls along the promenade.
Each stall, ***i***, has a distinct *flavour rating* (an integer between **1** and ***M***, inclusive) that reflects how popular its signature dish is.

You are asked to assign to each stall a positive integer number of *ticket-packs* ***t[i]*** (each pack is an identical block of customers) to send to that stall for the opening day.
Each ***t[i]*** must be at most ***M***.

Your goal is to choose ticket-packs so that:

1. The overall **weighted average rating** of the stalls (weighted by their ticket-packs) is an integer, i.e.

$$\frac{\sum_{i=1}^{N} r[i] \cdot t[i]}{\sum_{i=1}^{N} t[i]}$$

2. No three stalls in **increasing** stall-index order may receive exactly the same number of ticketpacks, that is, there must **not** exist indices $i < j < k$ with $t[i] = t[j] = t[k]$.
3. Every ***t[i]*** is an integer satisfying $1 \le t[i] \le M$.

If such an assignment is impossible, output **-1**.

Input:

The input consists of multiple test cases.

- The first line contains an integer T $(1 \le T \le 1000)$ which is the number of test cases.
- For each test case:
  - The first line contains two space-separated integers $N(1 \le N \le 10^6)$ and $M(1 \le M \le 10^9)$.
  - The second line contains ***N*** distinct integers $r[1], r[2], \ldots, r[N]$ $(1 \le r[i] \le M)$ denoting the flavour ratings.
- It is guaranteed that the sum of ***N*** across all test cases is at most 10^6.

Output:

For each test case, output one line:

- If an assignment exists, print ***N*** space-separated integers $t[1], t[2], \ldots, t[N]$ $(1 \le t[i] \le M)$ representing a valid ticket-pack assignment (any valid answer will be accepted).
- Otherwise, print **-1**.

Note: Sample test case is on the next page.

Sample Input	Sample Output
3 3 6 4 1 3 2 2 1 2 4 100000 1 2 5 9	2 1 2 -1 1 1 3 4

Explanation of Samples:

Case 1:

r = ***[4, 1, 3]***, *t* = ***[2, 1, 2]***

Weighted sum = $4 \cdot 2 + 1 \cdot 2 + 3 \cdot 2 = 15$

Total packs = 5

Weighted Average = 15/5 = 3 (an integer). No triple of equal ***t*** exists.

Case 2:

With ***N*** = ***2***, ***M*** = ***2,*** and ***r*** = ***[1, 2]***, none of the possible combinations of ***t[i]*** $\in$ ***{1, 2}*** make the average an integer, hence **-1**.

Case 3:

r = ***[1, 2, 5, 9]***, ***t*** = ***[1, 1, 3, 4]***

Weighted sum = $1 \cdot 1 + 2 \cdot 1 + 5 \cdot 3 + 9 \cdot 4 = 54$

Total packs = 9

Weighted Average = 54/9 = 6 (an integer). No triple of equal ***t*** exists.